A satellite imaging system captures a continuous image of an agricultural field. To process the image using a computer, the continuous image must be converted into a discrete digital image. The imaging system samples the field into a grid of 512 rows and 512 columns. Each sampled point becomes one pixel in the digital image. Apply the concept of discrete image representation and find the total number of pixels in the image.

A. 262,144 pixels
B. 131,072 pixels
C. 512 pixels
D. 1,024 pixels

Correct Answer: A. 262,144 pixels

Explanation:

$$Total\ Pixels = Rows \times Columns$$
$$Total\ Pixels = 512 \times 512 = 262144$$

So, the digital image contains 262,144 pixels.

A medical image scanner converts continuous brightness values into discrete intensity levels. The scanner uses 8 bits per pixel to represent each grayscale pixel. This means each pixel can store a fixed number of intensity values. The radiology team wants to know how many gray levels are available for representing the image. Apply the quantization formula and find the number of gray levels.

A. 128 levels
B. 256 levels
C. 512 levels
D. 1024 levels

Correct Answer: B. 256 levels

Explanation:

$$Gray\ Levels = 2^b$$

where $b = 8$.

$$Gray\ Levels = 2^8 = 256$$

So, an 8-bit grayscale image has 256 gray levels, from 0 to 255.

A document scanning system stores scanned pages as grayscale digital images. One scanned page has a resolution of 1024 × 768 pixels. Each pixel is represented using 8 bits, which is equal to 1 byte. The system administrator wants to calculate the storage needed for one uncompressed image. Apply digital image storage calculation and find the required storage.

A. 512 KB
B. 768 KB
C. 1024 KB
D. 2048 KB

Correct Answer: B. 768 KB

Explanation:

$$Total\ Pixels = 1024 \times 768 = 786432$$

Each pixel = 8 bits = 1 byte.

$$Storage = 786432 \text{ bytes}$$
$$786432 \div 1024 = 768 \text{ KB}$$

So, the storage required is 768 KB.

A smart classroom camera captures a color image of the board. The image size is 800 × 600 pixels. Each pixel is represented using three color channels: Red, Green, and Blue. Each channel uses 8 bits, so each RGB pixel requires 24 bits, or 3 bytes. Apply RGB image representation and find the total uncompressed storage required.

A. 1.44 MB
B. 2.40 MB
C. 3.60 MB
D. 4.80 MB

Correct Answer: A. 1.44 MB

Explanation:

$$Total\ Pixels = 800 \times 600 = 480000$$

Each RGB pixel = 3 bytes.

$$Storage = 480000 \times 3 = 1440000 \text{ bytes}$$
$$1440000 \text{ bytes} \approx 1.44 \text{ MB}$$

So, the storage required is 1.44 MB.

A robot vision system uses a discrete image grid to locate an object. The object center is detected at row 120 and column 200 in the image. In digital image representation, the row indicates the vertical position and the column indicates the horizontal position. The system must represent the object center using image coordinates. Apply pixel coordinate representation and identify the correct coordinate form.

A. $(x, y) = (120,200)$
B. $(x, y) = (200,120)$
C. $(x, y) = (320,120)$
D. $(x, y) = (120,320)$

Correct Answer: B. $(x, y) = (200,120)$

Explanation:
In image coordinates:

- Column represents x
- Row represents y

Given:

$$Row = 120, Column = 200$$

Therefore:

$$(x, y) = (200,120)$$

So, the correct coordinate is $(200, 120)$.

A camera system captures a continuous brightness value and converts it into an 8-bit grayscale value. The measured brightness is normalized to 0.6 on a scale from 0 to 1. In 8-bit image representation, the intensity range is from 0 to 255. The system converts the normalized brightness into a discrete pixel value. Apply intensity scaling and find the corresponding grayscale value.

A. 102
B. 128
C. 153
D. 180

Correct Answer: C. 153

Explanation:

$$Pixel\,Value = Normalized\,Value \times 255$$
$$Pixel\,Value = 0.6 \times 255 = 153$$

So, the grayscale value is 153.

A self-driving car uses edge detection to identify lane markings on the road. At one pixel location, the Sobel operator gives a horizontal gradient value $G_x = 6$ and a vertical gradient value $G_y = 8$. The system calculates the gradient magnitude to decide whether the pixel belongs to a strong edge. A higher magnitude indicates a stronger edge. Apply the gradient magnitude formula and find the edge strength.

A. 8
B. 10
C. 12
D. 14

Correct Answer: B. 10

Explanation:

$$G = \sqrt{G_x^2 + G_y^2}$$
$$G = \sqrt{6^2 + 8^2}$$
$$G = \sqrt{36 + 64} = \sqrt{100} = 10$$

So, the edge strength is 10.

A document scanning system uses edge detection to identify the boundary of printed text. After applying a gradient operator, one pixel has gradient magnitude 75. The system uses a threshold value of 60. If the gradient magnitude is greater than or equal to the threshold, the pixel is marked as an edge. Apply the thresholding rule and identify the output.

A. Edge pixel
B. Non-edge pixel
C. Background pixel only
D. Blurred pixel

Correct Answer: A. Edge pixel

Explanation:

Given:

$$Gradient = 75, Threshold = 60$$

Since:

$$75 \geq 60$$

The pixel is classified as an edge pixel.

A machine vision system inspects small cracks on a metal surface. For a 2×2image region, the pixel values are:

$$\begin{bmatrix} 100 & 120 \\ 130 & 160 \end{bmatrix}$$

The Roberts Cross approximation uses:

$$G_x = I(x, y) - I(x + 1, y + 1)$$

$$G_y = I(x+1, y) - I(x, y+1)$$

The system calculates the absolute gradient values to detect sharp changes. Apply the Roberts operator and find $|G_x| + |G_y|$.

A. 50
B. 60
C. 70
D. 90

Correct Answer: C. 70

Explanation:

$$G_x = 100 - 160 = -60$$
$$|G_x| = 60$$
$$G_y = 130 - 120 = 10$$
$$|G_y| = 10$$
$$|G_x| + |G_y| = 60 + 10 = 70$$

So, the edge response is 70.

A surveillance camera detects object boundaries in a grayscale image. At a particular pixel, the Prewitt operator gives $G_x = 15$and $G_y = 20$. The system calculates the gradient magnitude to determine whether the boundary is strong enough for object segmentation. Apply the gradient magnitude formula and calculate the edge magnitude.

A. 20
B. 25
C. 30
D. 35

Correct Answer: B. 25

Explanation:

$$G = \sqrt{G_x^2 + G_y^2}$$
$$G = \sqrt{15^2 + 20^2}$$
$$G = \sqrt{225 + 400}$$
$$G = \sqrt{625} = 25$$

So, the edge magnitude is 25.

A robot vision system checks whether there is a sudden intensity change between two neighboring pixels. Pixel A has intensity 45, and Pixel B has intensity 110. The system uses the absolute intensity difference between neighboring pixels to identify possible edges. If the difference is high, it indicates a boundary. Apply the intensity difference method and calculate the edge response.

A. 45
B. 55
C. 65
D. 75

Correct Answer: C. 65

Explanation:

$$Difference = |\ 110 - 45\ |$$
$$Difference = 65$$

So, the edge response is 65.

A medical image system uses Canny edge detection to find boundaries of blood vessels. The lower threshold is 40, and the higher threshold is 90. A pixel has gradient magnitude 95. In Canny edge detection, pixels above the higher threshold are treated as strong edges. Apply the Canny threshold rule and classify the pixel.

A. Strong edge
B. Weak edge
C. Non-edge
D. Noise only

Correct Answer: A. Strong edge

Explanation:

Given:

$$Gradient = 95, High\ Threshold = 90$$

Since:

$$95 > 90$$

The pixel is classified as a strong edge.

1. Applying Feature Vector Representation

A face recognition system extracts two simple features from a face image: eye distance and nose width. For one person, the eye distance is 60 pixels, and the nose width is 25 pixels. The system stores these values as a feature vector for comparison with other faces. This feature vector helps the system recognize the person later. Apply the concept of feature extraction and identify the correct feature vector.

A. $(60, 25)$
B. $(25, 60)$
C. $(60 + 25, 60 - 25)$
D. (60×25)

Correct Answer: A. $(60, 25)$

Explanation:
A feature vector stores extracted feature values in an ordered form.

$$Feature\ Vector = (Eye\ Distance,\ Nose\ Width)$$
$$Feature\ Vector = (60{,}25)$$

So, the correct feature vector is $(60, 25)$.

A road sign detection system extracts edge features from a traffic sign image. After applying an edge detector, the system finds 850 edge pixels in the image. Out of these, 120 pixels are removed as noise. The remaining edge pixels are used as shape features for recognition. Apply feature selection and find the number of useful edge features.

A. 620
B. 700
C. 730
D. 850

Correct Answer: C. 730

Explanation:

$$Useful\ Edge\ Features = Total\ Edge\ Pixels - Noise\ Pixels$$
$$Useful\ Edge\ Features = 850 - 120 = 730$$

So, 730 edge pixels are used as useful features.

A content-based image retrieval system extracts color histogram features from an image. In a small image region, there are 200 red pixels, 150 green pixels, and 50 blue pixels. The system wants to calculate the total number of color feature counts used in the histogram. Apply histogram feature extraction and find the total count.

A. 300
B. 350
C. 400
D. 450

Correct Answer: C. 400

Explanation:

$$Total\ Count = Red + Green + Blue$$
$$Total\ Count = 200 + 150 + 50 = 400$$

So, the total histogram count is 400.

A medical image analysis system extracts texture features from a tissue image. Four neighboring gray-level values in a region are 40, 50, 60, and 70. The system calculates the average intensity as a simple texture feature. This helps compare normal and abnormal tissue regions. Apply average intensity calculation and find the texture feature value.

A. 45
B. 50
C. 55
D. 60

Correct Answer: C. 55

Explanation:

$$Average = \frac{40 + 50 + 60 + 70}{4}$$
$$Average = \frac{220}{4} = 55$$

So, the texture feature value is 55.

An object recognition system detects a rectangular object in an image. The bounding box around the object has width 120 pixels and height 60 pixels. The system calculates aspect ratio as a shape feature to help classify the object. Different objects may have different aspect ratio patterns. Apply the aspect ratio formula and find the value.

A. 0.5
B. 1.0
C. 1.5
D. 2.0

Correct Answer: D. 2.0

Explanation:

$$Aspect\ Ratio = \frac{Width}{Height}$$
$$Aspect\ Ratio = \frac{120}{60} = 2.0$$

So, the aspect ratio is 2.0.

A machine learning model is trained to classify medical images as normal or abnormal. The model gives a training loss of 0.40. To avoid overfitting, L2 regularization is added. The regularization parameter is $\lambda = 0.1$, and the sum of squared weights is 5. Apply L2 regularization and find the final loss.

A. 0.50
B. 0.70
C. 0.90
D. 1.20

Correct Answer: C. 0.90

Explanation:

$$Final\ Loss = Training\ Loss + \lambda \sum w^2$$

$$Final\ Loss = 0.40 + 0.1 \times 5$$
$$Final\ Loss = 0.40 + 0.50 = 0.90$$

So, the final loss is 0.90.

2. Applying L1 Regularization Penalty

A computer vision model uses L1 regularization to reduce unnecessary features. The model has four weights: $w = [2, -3, 1, -4]$. Apply L1 regularization and calculate the penalty value.

A. 6
B. 8
C. 10
D. 12

Correct Answer: C. 10

Explanation:

The L1 penalty is calculated as the sum of absolute weight values:

$$L1 = \sum |w|$$

$$L1 = |2| + |-3| + |1| + |-4|$$
$$L1 = 2 + 3 + 1 + 4 = 10$$

So, the L1 penalty is 10.

An object recognition model uses L2 regularization to control large weight values. The model has three weights: $w = [3,4,2]$. Apply L2 regularization and find the penalty value.

A. 19
B. 25
C. 29
D. 36

Correct Answer: C. 29

Explanation:

The L2 penalty is calculated as:

$$L2 = \sum w^2$$

$$L2 = 3^2 + 4^2 + 2^2$$
$$L2 = 9 + 16 + 4 = 29$$

So, the L2 penalty is 29.

A deep learning model for face recognition has a training error of 0.25. The model applies regularization to avoid memorizing the training images. The penalty term is 0.15. Apply the regularized cost function and find the total cost.

A. 0.10
B. 0.25
C. 0.30
D. 0.40

Correct Answer: D. 0.40

Explanation:

The total cost is calculated as:

$$Total\ Cost = Training\ Error + Regularization\ Penalty$$

$$Total\ Cost = 0.25 + 0.15$$
$$Total\ Cost = 0.40$$

So, the total cost is 0.40.

A model uses weight decay during training. The current weight value is 0.80, the learning rate is 0.1, and the regularization parameter is $\lambda = 0.5$. Apply the weight decay formula and find the new weight.

A. 0.76
B. 0.70
C. 0.60
D. 0.40

Correct Answer: A. 0.76

Explanation:

The weight decay update is calculated as:

$$w_{new} = w - \eta\lambda w$$

where $\eta$is the learning rate.

$$w_{new} = 0.80 - (0.1 \times 0.5 \times 0.80)$$
$$w_{new} = 0.80 - 0.04$$
$$w_{new} = 0.76$$

So, the new weight is 0.76.

6. Applying Overfitting Check Using Accuracy Gap

A computer vision classifier is trained to detect defects in industrial products. The training accuracy is 98%, but the testing accuracy is only 78%. The large difference indicates that the model performs very well on training data but poorly on unseen data. Regularization is suggested to reduce this problem. Apply the concept of overfitting and find the accuracy gap.

A. 10%
B. 15%
C. 20%
D. 25%

Correct Answer: C. 20%

Explanation:

$$Accuracy\ Gap = Training\ Accuracy - Testing\ Accuracy$$
$$Accuracy\ Gap = 98\% - 78\% = 20\%$$

So, the accuracy gap is 20%.

A traffic monitoring camera captures two consecutive frames of a moving car. In frame 1, a corner point on the car is located at $(40, 60)$. In frame 2, the same point moves to $(48, 66)$. The optical flow system calculates the displacement of the point between frames to estimate motion direction and speed. Apply the optical flow displacement concept and find the displacement vector.

A. $(6, 8)$
B. $(8, 6)$
C. $(48, 66)$
D. $(40, 60)$

Correct Answer: B. $(8, 6)$

Explanation:

$$u = x_2 - x_1 = 48 - 40 = 8$$
$$v = y_2 - y_1 = 66 - 60 = 6$$

So, the optical flow displacement vector is:

$$(u, v) = (8,6)$$

A sports analytics system tracks the motion of a football between two video frames. The optical flow vector of the ball is calculated as $(u, v) = (3,4)$. The system wants to compute the magnitude of motion to estimate how strongly the ball has moved between frames. Apply the optical flow magnitude formula and find the motion magnitude.

A. 3 pixels
B. 4 pixels
C. 5 pixels
D. 7 pixels

Correct Answer: C. 5 pixels

Explanation:

$$Magnitude = \sqrt{u^2 + v^2}$$
$$Magnitude = \sqrt{3^2 + 4^2}$$
$$Magnitude = \sqrt{9 + 16} = \sqrt{25} = 5$$

So, the motion magnitude is 5 pixels.

A drone camera tracks an object moving across a field. The optical flow vector for one feature point is $(u, v) = (6,8)$. Apply the direction formula and find the approximate angle.

A. $30°$
B. $45°$
C. $53°$
D. $60°$

Correct Answer: C. $53°$

Explanation:

The system calculates the motion direction angle using:

$$\theta = \tan^{-1}\left(\frac{v}{u}\right)$$

This helps identify the movement direction of the object.

$$\theta = \tan^{-1}\left(\frac{8}{6}\right)$$
$$\theta = \tan^{-1}(1.33)$$
$$\theta \approx 53^\circ$$

So, the approximate direction angle is 53°.

A surveillance system observes a person walking inside a corridor. A detected feature point moves 12 pixels between two consecutive frames. The video frame interval is 0.5 seconds. The system estimates the pixel velocity of the moving point. Apply the velocity formula and calculate the motion speed.

A. 12 pixels/s
B. 18 pixels/s
C. 20 pixels/s
D. 24 pixels/s

Correct Answer: D. 24 pixels/s

Explanation:

$$Velocity = \frac{Displacement}{Time}$$
$$Velocity = \frac{12}{0.5} = 24 \text{ pixels/s}$$

So, the motion speed is 24 pixels/s.

An optical flow method assumes brightness constancy between consecutive frames. At a pixel, the image gradients are $I_x = 2$, $I_y = 3$, and temporal gradient $I_t = -12$. The horizontal flow component is $u = 3$. Apply the optical flow constraint equation and find v.

A. 1
B. 2
C. 3
D. 4

Correct Answer: B. 2

Explanation:

The brightness constancy equation is:

$$I_x u + I_y v + I_t = 0$$

The system must calculate the vertical flow component v.

$$\begin{aligned}
&I_x u + I_y v + I_t = 0 \\
&2(3) + 3v - 12 = 0 \\
&6 + 3v - 12 = 0 \\
&3v = 6 \\
&v = 2
\end{aligned}$$

So, the vertical flow component is 2.

A video processing system classifies movement based on optical flow magnitude. If the magnitude is below 2 pixels, it is considered slow motion. If it is between 2 and 8 pixels, it is considered moderate motion. If it is above 8 pixels, it is considered fast motion. A detected object has optical flow vector $(u, v) = (6,8)$. Apply the motion classification rule and identify the motion type.

A. Slow motion
B. Moderate motion
C. Fast motion
D. No motion

Correct Answer: C. Fast motion

Explanation:

$$\begin{aligned}
&Magnitude = \sqrt{6^2 + 8^2} \\
&Magnitude = \sqrt{36 + 64} = \sqrt{100} = 10
\end{aligned}$$

Since $10 > 8$, the object is classified as fast motion.

A warehouse robot uses two cameras to estimate the distance of boxes placed on a shelf. A point on a box appears at horizontal coordinate 240 pixels in the left camera image and 200 pixels in the right camera image. The stereo vision system calculates disparity before estimating depth. Disparity helps the robot understand how close or far the object is. Apply the disparity formula and find the disparity value.

A. 20 pixels
B. 30 pixels
C. 40 pixels
D. 50 pixels

Correct Answer: C. 40 pixels

Explanation:

$$d = x_L - x_R$$
$$d = 240 - 200 = 40$$

So, the disparity is 40 pixels.

An autonomous vehicle uses stereo cameras to estimate the distance of a pedestrian. The baseline between the two cameras is 0.2 m, the focal length is 800 pixels, and the disparity of the pedestrian point is 40 pixels. The system uses this depth value to decide whether to slow down. Apply the stereo depth formula and calculate the depth.

A. 2 m
B. 3 m
C. 4 m
D. 5 m

Correct Answer: C. 4 m

Explanation:

$$Z = \frac{fB}{d}$$

where $f = 800$, $B = 0.2$, and $d = 40$.

$$Z = \frac{800 \times 0.2}{40}$$
$$Z = \frac{160}{40} = 4 \text{ m}$$

So, the depth of the pedestrian is 4 m.

A mobile robot moves inside a laboratory and uses stereo vision to avoid obstacles. The stereo camera has a baseline of 0.1 m and focal length of 600 pixels. An obstacle point has a disparity of 30 pixels. The robot calculates the obstacle distance before planning its path. Apply the stereo vision depth equation and find the distance.

A. 1 m
B. 2 m
C. 3 m
D. 4 m

Correct Answer: B. 2 m

Explanation:

$$Z = \frac{fB}{d}$$

$$Z = \frac{600 \times 0.1}{30}$$
$$Z = \frac{60}{30} = 2 \text{ m}$$

So, the obstacle is at a distance of 2 m.

A stereo camera system in a drone observes two objects. Object A has a disparity of 50 pixels, and Object B has a disparity of 25 pixels. The focal length and camera baseline are the same for both objects. The drone must identify which object is closer based on disparity. Apply the relationship between disparity and depth and choose the correct statement.

A. Object A is closer because it has larger disparity
B. Object B is closer because it has smaller disparity
C. Both objects are at the same distance
D. Disparity is not related to depth

Correct Answer: A. Object A is closer because it has larger disparity

Explanation:

In stereo vision:

$$Z = \frac{fB}{d}$$

Depth $Z$is inversely proportional to disparity d. Therefore, larger disparity means smaller depth, so the object is closer.

A stereo camera setup is used in an industrial inspection robot. The baseline between the two cameras is 0.3 m, the focal length is 1000 pixels, and the measured disparity is 60 pixels. The system calculates the depth of a component on the conveyor belt. Apply the stereo depth formula and find the depth.

A. 3 m
B. 4 m
C. 5 m
D. 6 m

Correct Answer: C. 5 m

Explanation:

$$Z = \frac{fB}{d}$$
$$Z = \frac{1000 \times 0.3}{60}$$
$$Z = \frac{300}{60} = 5 \text{ m}$$

So, the component is 5 m away.

A stereo vision system estimates the 3D location of a point in front of a robot. The depth of the point is calculated as 4 m. The image coordinate is $x = 200$pixels, the focal length is $f = 800$pixels, and the principal point is assumed to be zero for simplicity. Apply the formula and calculate the Xcoordinate.

A. 0.5 m
B. 1.0 m
C. 1.5 m
D. 2.0 m

Correct Answer: B. 1.0 m

Explanation:

The system estimates the real-world Xcoordinate using:

$$X = \frac{xZ}{f}$$

$$X = \frac{xZ}{f}$$
$$X = \frac{200 \times 4}{800}$$
$$X = \frac{800}{800} = 1.0 \text{ m}$$

So, the Xcoordinate is 1.0 m.

A traffic surveillance camera captures two consecutive frames of a moving car. In frame 1, a feature point on the car is located at $(30, 40)$. In frame 2, the same feature point moves to $(42, 49)$. The system estimates motion by calculating the displacement between the two frames. Apply the motion estimation concept and find the motion vector.

A. $(9, 12)$
B. $(12, 9)$
C. $(42, 49)$
D. $(30, 40)$

Correct Answer: B. $(12, 9)$

Explanation:

$$u = x_2 - x_1 = 42 - 30 = 12$$
$$v = y_2 - y_1 = 49 - 40 = 9$$

So, the motion vector is:

$$(u, v) = (12,9)$$

A drone camera monitors a moving vehicle from above. The vehicle motion vector between two frames is $(u, v) = (3,4)$. This direction helps the drone understand the movement path of the vehicle. Apply the direction formula and find the approximate angle.

A. $37°$
B. $45°$
C. $53°$
D. $60°$

Correct Answer: C. $53°$

Explanation:

The system estimates the motion direction using:

$$\theta = \tan^{-1}\left(\frac{v}{u}\right)$$

$$\theta = \tan^{-1}\left(\frac{4}{3}\right)$$
$$\theta \approx 53°$$

So, the approximate motion direction is $53°$.

A CCTV system observes a person walking in a corridor. A detected point on the person moves 18 pixels between two frames. The time difference between the two frames is 0.6 seconds. The system calculates pixel velocity to estimate the walking speed in the image. Apply the velocity formula and find the pixel velocity.

A. 20 pixels/s
B. 25 pixels/s
C. 30 pixels/s
D. 35 pixels/s

Correct Answer: C. 30 pixels/s

Explanation:

$$Velocity = \frac{Displacement}{Time}$$
$$Velocity = \frac{18}{0.6} = 30 \text{ pixels/s}$$

So, the pixel velocity is 30 pixels/s.

A video compression system uses block matching for motion estimation. A block is located at coordinate $(50,70)$in the current frame. The best matching block in the previous frame is located at $(44,65)$. The motion vector is calculated as the difference between the current block position and the matched block position. Apply block-based motion estimation and find the motion vector.

A. $(6,5)$
B. $(5,6)$
C. $(44,65)$
D. $(50,70)$

Correct Answer: A. $(6,5)$

Explanation:

$$u = 50 - 44 = 6$$
$$v = 70 - 65 = 5$$

So, the motion vector is:

$$(u, v) = (6,5)$$

A video processing system compares two small image blocks to find the best motion match. The absolute pixel differences between the current block and a candidate block are: $4,\ 6,\ 8,\ 2$
The system uses Mean Absolute Difference, or MAD, to measure block similarity. Lower MAD indicates a better match. Apply the MAD formula and calculate the value.

A. 4
B. 5
C. 6
D. 8

Correct Answer: B. 5

Explanation:

$$MAD = \frac{4 + 6 + 8 + 2}{4}$$
$$MAD = \frac{20}{4} = 5$$

So, the MAD value is 5.

A drone captures multiple images of a college building from different viewpoints. The Structure from Motion system detects 500 feature points in one image. Out of these, 380 feature points are correctly matched with the next image. These matched features are used to estimate camera motion and reconstruct 3D points. Apply the feature matching concept and find the percentage of correctly matched features.

A. 70%
B. 76%
C. 80%
D. 85%

Correct Answer: B. 76%

Explanation:

$$Matching\ Percentage = \frac{Correct\ Matches}{Total\ Features} \times 100$$
$$Matching\ Percentage = \frac{380}{500} \times 100 = 76\%$$

So, the correctly matched feature percentage is 76%.

A heritage preservation team uses Structure from Motion to create a 3D model of an old temple. The software identifies 1200 matched feature points across multiple images. After triangulation, only 900 points are successfully reconstructed as valid 3D points. The remaining points are removed because of poor matching or geometric error. Apply the reconstruction percentage formula and find the percentage of valid 3D points.

A. 65%
B. 70%
C. 75%
D. 80%

Correct Answer: C. 75%

Explanation:

$$Valid\ Reconstruction\ Percentage = \frac{Valid\ 3D\ Points}{Matched\ Feature\ Points} \times 100$$
$$= \frac{900}{1200} \times 100 = 75\%$$

So, the valid 3D reconstruction percentage is 75%.

A Structure from Motion system reconstructs a 3D point and projects it back onto an image. The observed feature point in the image is located at $(150, 200)$. The reprojected point from the reconstructed 3D model is located at $(154, 203)$. The system calculates reprojection error to check reconstruction accuracy. Apply the reprojection error formula and calculate the error.

A. 3 pixels
B. 4 pixels
C. 5 pixels
D. 7 pixels

Correct Answer: C. 5 pixels

Explanation:

$$Error = \sqrt{(154 - 150)^2 + (203 - 200)^2}$$
$$Error = \sqrt{4^2 + 3^2}$$
$$Error = \sqrt{16 + 9} = 5$$

So, the reprojection error is 5 pixels.

A 3D reconstruction system captures 6 images of a statue from different viewpoints. To perform pairwise feature matching, the system compares every image with every other image once. The engineer wants to calculate the total number of unique image pairs used for matching. Apply the image pair formula and find the number of unique pairs.

A. 10 pairs
B. 12 pairs
C. 15 pairs
D. 18 pairs

Correct Answer: C. 15 pairs

Explanation:

For $n$images, number of unique image pairs is:

$$\frac{n(n-1)}{2}$$
$$\frac{6(6-1)}{2} = \frac{6 \times 5}{2} = 15$$

So, the system forms 15 unique image pairs.

A Structure from Motion system reconstructs a building model. In the reconstructed model, the height of a door is measured as 2 units. The actual height of the door is 2 meters. The system uses this known object to estimate scale. A wall in the reconstructed model has height 5 units. Apply the scale factor and find the real height of the wall.

A. 3 m
B. 4 m
C. 5 m
D. 6 m

Correct Answer: C. 5 m

Explanation:

Given:

$$2 \text{ units} = 2 \text{ meters}$$

So,

$$1 \text{ unit} = 1 \text{ meter}$$

Wall height:

$$5 \text{ units} = 5 \text{ meters}$$

So, the real height of the wall is 5 m.